

Same Theorem, Different Interior

Human and AI Proofs as Alternative Epistemic Architectures

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Abstract

How much of a proof’s epistemic structure is fixed by what it proves, and how much is a consequence of who or what produced it? Viteri and DeDeo argued that mathematical confidence can arise from the network organization of proofs: converging deductive and abductive pathways can overcome local fallibility. Their evidence compared different theorems represented at very different levels of resolution. We instead hold the formal theorem fixed. We compare 2,321 paired human- and AI-provenance Lean scripts and, in a separately filtered sample, 312 pairs of elaborated proof-term networks. We distinguish a proof’s *interface*—its contact with named library declarations—from its *interior* of local claims and expression structure.

The theorem strongly constrains proof scale: human and AI line counts correlate at $\rho_s = 0.67$ and elaborated node counts at $\rho_s = 0.76$. Against this shared constraint, AI scripts contain 1.41 times as many local claims per tactic, and AI terms contain 1.12 times as many internal expression nodes per distinct constant. Among the 184 pairs measured both ways, the two interior contrasts correlate at $\rho_s = 0.19$. Yet source-level premise differences have paired median zero, elaborated terms contain essentially the same number of distinct constants, and the Viteri–DeDeo belief model assigns nearly identical theorem confidence to the two populations. Across three local-error settings, the mean paired belief difference is 0.00012 (95% source-cluster interval [-0.0019, 0.0020]), while the association between interior expansion and belief change is only $\rho_s = 0.046$.

We call this pattern *epistemic decoupling*: alternative proofs can reorganize substantial internal derivational work without proportionately changing their external mathematical interface or modeled epistemic endpoint. The result shifts the study of AI proofs from correctness alone to the design of interfaces between certificates, readers, and mathematical libraries.

1 Introduction

A proof is at once a derivation, an argument, and an artifact for future use. These roles coincide only imperfectly. A kernel-checked term can establish formal derivability without showing a human reader the main idea; a lucid argument can omit routine steps; a library proof can be correct yet difficult to maintain or reuse. The distinction has become practically urgent because AI systems now produce large numbers of verified formal proofs (Yang et al., 2023; Ren et al., 2025; Hubert et al., 2026). Once correct certificates become abundant, the central question is no longer only whether a theorem can be proved, but what kind of proof has been produced and how it enters mathematical knowledge.

Viteri and DeDeo (2022) gave a network answer to a prior version of this problem. If each deductive step can fail, a long chain should become less trustworthy as it grows. Real proofs, however, are not chains. Claims are reused, lines of reasoning converge, and conclusions can provide abductive support for their premises. In the resulting network, the exponential loss of influence along

one path can be offset by the proliferation of paths. Mathematical confidence can then undergo an epistemic phase transition from local uncertainty to global conviction. This account replaces an ideal reasoner with a finite, fallible one and connects formal dependency structure to cognitive theories of coherence and plausible inference (Pólya, 1954; Avigad, 2022).

The original evidence established a striking commonality across historical and machine-assisted proof networks. It did not isolate the role of the theorem from the role of the proof producer. Euclid, Wiles, and a Coq formalization differ simultaneously in mathematical content, author, purpose, language, granularity, and extraction procedure. The growth of formal proof datasets now makes a more controlled comparison possible: two valid proofs of the *same formal statement*, one with human provenance and one with AI provenance.

This paired design motivates a distinction between a proof’s *interface* and its *interior*. The interface consists of the named mathematical declarations through which the proof contacts a library. The interior is the local derivational structure that connects those declarations to the theorem. Two proofs may expose a similar interface while organizing their interiors differently. That possibility matters because proof assistants verify the entire interior, whereas human readers usually navigate a compressed presentation through named lemmas, goals, examples, and trusted background (Weber, 2008; Inglis and Alcock, 2012; Panse et al., 2018).

We test three propositions. First, the theorem should constrain proof architecture: paired human and AI proofs should resemble one another despite their different provenance. Second, production system should matter within that constraint: present AI systems may insert more internal scaffolding without drawing on a proportionately broader named interface. Third, if the redundant network mechanisms identified by Viteri and DeDeo are robust, additional interior structure need not lower theorem-level confidence. Together these propositions define a novel thesis: *epistemic architecture can vary substantially while epistemic outcome remains stable*.

2 Proofs as interface-bearing networks

2.1 Four objects that should not be collapsed

Let a formal theorem be a type S , and let a checked proof term P inhabit S . From P we can extract a directed acyclic expression graph $G(P)$ in which repeated subexpressions are shared. The source script $R(P)$ is a further object: a sequence of tactics and local declarations that caused an elaborator to construct P . Finally, a mathematical reader encounters a presentation $M(P)$, which may be the source script, a prose proof, an explanation, or some transformation of the certificate. Kernel correctness is a relation between P and S ; comprehension and uptake are relations involving $M(P)$, a reader, and a community.

The graph $G(P)$ contains a natural operational boundary. Constant nodes refer to named declarations in the ambient library. Treat these as the proof’s interface $C(P)$. The remaining nodes implement applications, abstractions, types, variables, and other expression structure. We define the proof’s *interior load* as

$$\chi(P) = \frac{|V(G(P))| - |C(P)|}{|C_{\text{distinct}}(P)| + 1}. \quad (1)$$

This is not a measure of mathematical sophistication. It asks how much elaborated structure is placed behind each distinct named point of contact with the library. At source level, a parallel proxy is the number of explicit local `have` claims per tactic invocation.

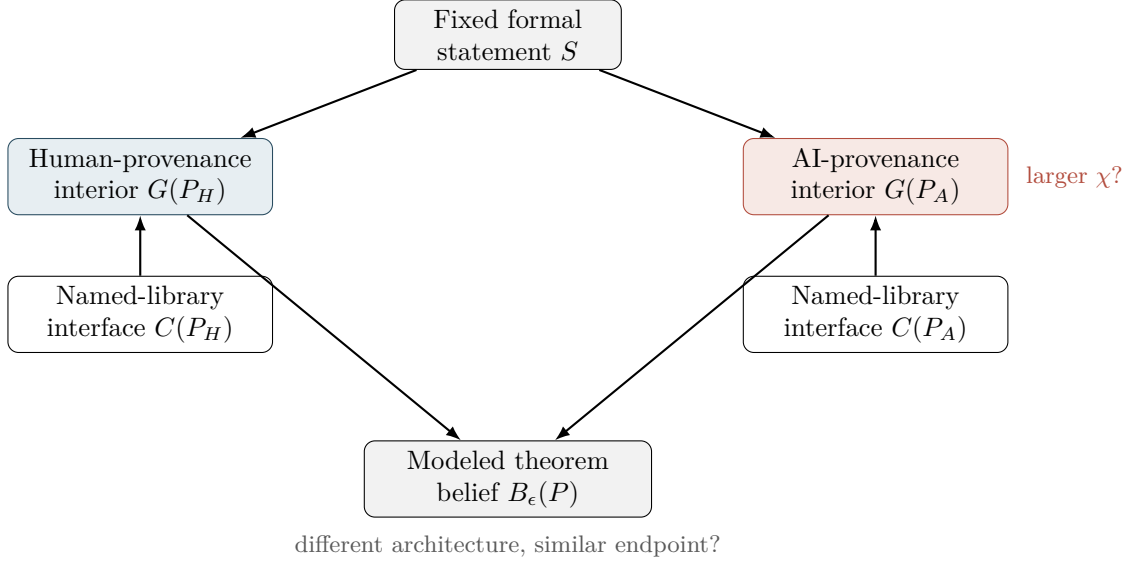


Figure 1: The paired design holds the formal statement fixed and separates the named-library interface from the proof’s internal expression structure. Epistemic decoupling occurs when the interior changes without a commensurate change in modeled theorem belief.

2.2 From phase transition to representational robustness

In the Viteri–DeDeo model, nodes carry fallible binary beliefs and adjacent claims exert deductive and abductive influence. A graph can support high theorem confidence when evidentiary influence has enough mutually reinforcing routes. The present paired design asks a question the historical comparison could not: is that confidence robust across alternative realizations of the same theorem?

There are two competing intuitions. The linear-chain intuition predicts that a larger interior creates more opportunities for error and should reduce confidence. The network intuition predicts that additional structure can preserve or increase confidence when it creates shared and convergent support. A third possibility is cognitive: elaborated expression nodes may not correspond to the units on which human readers reason at all. In that case, term-level network differences can be real while human epistemic differences depend mostly on how the interface is presented and chunked. The empirical comparison below distinguishes the first two possibilities inside the model and turns the third into a behavioral research program.

3 Data and methods

3.1 Same-statement Lean scripts

The source analysis begins with NuminaMath-LEAN proof artifacts (AI-MO, 2025). Each record contains a single formal statement, a human-provenance Lean proof, an AI-provenance Lean proof, validation metadata, and a coarse problem-source label. We retain records for which both proofs are marked valid and available and both contain a measurable proof body with at least one premise-like reference. The resulting sample contains 2,321 theorem-matched pairs from 12 source groups.

For each script we count lines, tactic invocations, and explicit `have` declarations. We use two deliberately limited source-interface measures after removing comments and strings: a loose count of distinct dotted or capitalized identifiers and a strict count of distinct dotted identifiers. These

are lexical proxies, not authoritative dependency counts. Their purpose is to test whether a visible source-level contrast survives reasonable changes in the token rule.

To compare internal decomposition at source level while retaining zero-**have** proofs, we use

$$\chi_R = \frac{n_{\text{have}} + 1/2}{n_{\text{tactic}} + 1}. \quad (2)$$

The pseudocount affects the numeric ratio, so we also report raw medians and paired differences.

3.2 Elaborated proof-term networks

Lean elaboration converts source scripts into kernel-checkable terms. We extract the theorem type and value as a deduplicated expression DAG, directing edges from child expressions toward the expressions that contain them. Constants are left as named frontier nodes rather than recursively expanding all of Mathlib. This `term0` representation therefore captures the complete root term together with its named interface, not the transitive implementation of imported declarations.

Of 494 pairs successfully extracted on both sides, we exclude a pair if either graph contains `sorryAx` or `SyntheticOpaque`, labels indicating Lean error recovery rather than a completed proof. This leaves 312 clean pairs from the same 12 source groups. The exclusion is pairwise and substantial; the retained set should not be treated as a random sample of all valid proofs. The source and term samples are independently filtered; 184 pairs appear in both.

We measure total nodes, distinct constants, constant-node share, and the interior load $\chi(P)$ in [equation \(1\)](#). Total size includes both the theorem’s type and value. The paired design is therefore essential: an absolute node count is not interpretable as human proof length, but a within-statement contrast reveals how two terms realize the same formal target.

3.3 Belief propagation

For each clean term graph, we implement the symmetric-error case of the asymmetric Ising belief model in [Viteri and DeDeo \(2022\)](#). Nodes begin true with prior probability 0.75; deductive and abductive couplings correspond to one-step error rates $\epsilon \in \{.10, .05, .01\}$. We run ten sweeps and 32 Monte Carlo chains per graph, using common seeds within each human/AI pair to reduce avoidable simulation noise. Every retained graph is one weak component containing the theorem root, so theorem belief is not confounded by isolates in this sample. Our primary epistemic outcome is theorem-node belief, averaged over the three error settings. This is a consequence of a theoretical model, not a measurement of a human participant.

3.4 Paired estimands and source uncertainty

We report medians of paired differences or paired log ratios. Spearman correlations quantify how strongly theorem identity constrains scale across the two proof populations. To avoid treating thousands of competition problems as identically distributed independent observations, intervals resample the 12 coarse source groups with replacement, retaining all observations in a selected group. The final analysis uses 20,000 bootstrap resamples with a fixed seed. A second, identical run on the ORCHARD cluster reproduced the local Mac output byte for byte.

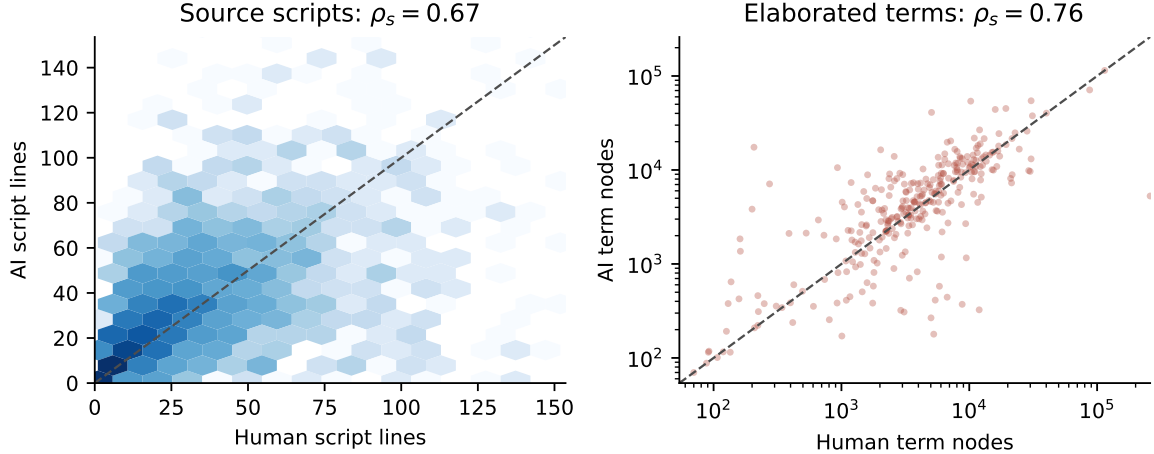


Figure 2: The same theorem strongly constrains proof scale. Each point is a theorem-matched human/AI pair; the dashed line is parity. The left panel uses a density representation because the source sample is large. Correlations are descriptive paired Spearman coefficients.

4 Results

4.1 The theorem is a strong architectural constraint

Human and AI proofs of the same statement are far from independent constructions. Script line counts have paired Spearman correlation $\rho_s = 0.67$ (95% source-cluster interval $[0.59, 0.74]$), and elaborated term node counts correlate at $\rho_s = 0.76$ (95% source-cluster interval $[0.67, 0.84]$). The association survives changes in source domain represented by the cluster intervals and is numerically higher after elaboration (figure 2).

This constraint is conceptually important. A human/AI difference is not a contrast between an organized human object and arbitrary machine text. Both proofs respond to the same statement, types, hypotheses, and ambient library. Much of their architecture is therefore problem-specific. Production system appears as a deformation around a shared theorem-imposed structure.

4.2 AI proofs expand the interior more than the interface

The deformation is asymmetric. AI scripts are only modestly longer: human and AI medians are 23 and 26 lines, with a median paired difference of +2 (95% source-cluster interval $[1, 3]$). They contain more explicit local decomposition, however. The median number of **have** claims is 3 for human scripts and 6 for AI scripts; the paired median difference is +1 ($[0, 1]$). Relative to tactic count, the median paired AI/human ratio is 1.41 (95% source-cluster interval $[1.27, 1.53]$). Giving each of the 12 source groups equal influence weakens the raw **have** contrast to +0.5 $[0, 1.5]$, showing that the tendency is heterogeneous rather than universal.

The source interface does not expand in parallel. The loose premise proxy has medians 5 and 4, and the strict proxy has medians 3 and 2, but both have median paired difference zero under the source-cluster bootstrap. The lower unpaired AI medians therefore do not imply that an AI proof of a given theorem uses fewer premises; the mass of pairwise ties and asymmetric differences matters. The zero-width median intervals reflect discreteness and repeated ties across only 12 coarse clusters; they are not formal evidence of exact equivalence.

Elaborated terms give a non-lexical version of the same pattern. AI terms have a median of

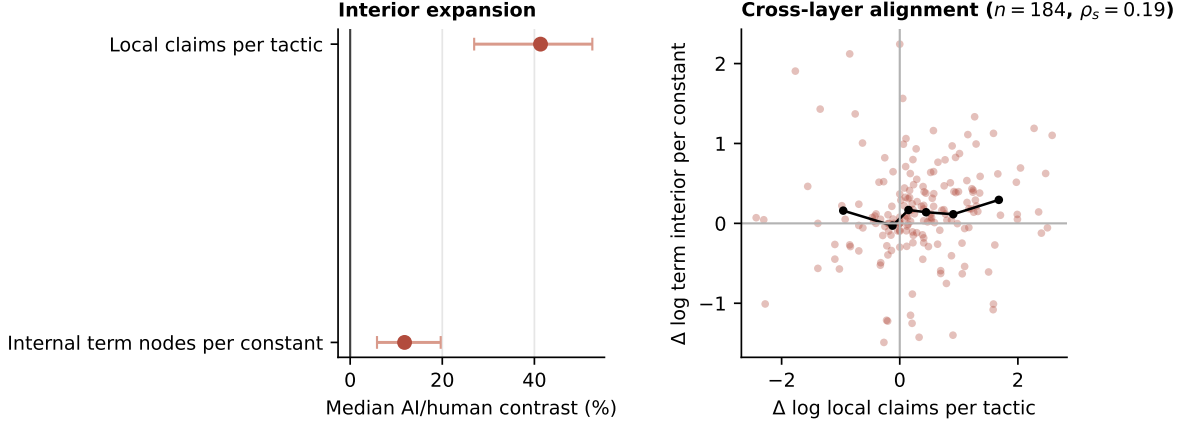


Figure 3: Two operationally independent measures of internal scaffolding. (a) Median paired AI/human ratios, expressed as percentage contrasts; intervals resample problem-source groups. (b) In the overlapping sample, source-level and term-level changes have a modest positive association. The black line joins six binned means and is descriptive.

4,561 nodes versus 3,816.5 for human terms, with a median paired difference of +290 ([20, 529]). But the number of distinct constants is almost unchanged: 200.5 versus 199, with a paired difference of -1 ($[-3, 0]$). Consequently, constants occupy a smaller fraction of AI terms, and the interior load is larger. AI terms contain 1.12 times as many non-constant expression nodes per distinct constant (95% source-cluster interval $[1.06, 1.20]$). Equal weighting of source groups gives a similar ratio of 1.11 ($[1.02, 1.21]$).

The two measures also align within proofs. Among the 184 theorem pairs observed at both levels, the paired change in local claims per tactic correlates with the paired change in term interior per constant at $\rho_s = 0.19$ (95% source-cluster interval $[0.04, 0.29]$; figure 3b). The association is modest, as expected when source tactics are elaborated through automation, but it makes the interior construct less dependent on either representation alone.

The conjunction, not any single count, motivates the interface-interior thesis. AI proofs do not simply scale every feature upward. They place more local and elaborated structure behind a named interface whose breadth changes little. This is compatible with explicit subgoal decomposition, coercion and normalization structure, repeated tactic templates, or less economical use of theorem APIs. The present data do not identify which mechanism dominates, and “more interior” should not be translated into “less understanding.”

4.3 Interior expansion does not produce a theorem-belief deficit

If every additional expression were another independent opportunity for failure, the larger AI interior should reduce theorem confidence. The network model gives a different answer. Aggregate human and AI belief profiles nearly overlap across all three local-error rates (figure 4a). Averaged across settings, the mean paired AI-minus-human theorem-belief difference is 0.00012, with 95% source-cluster interval $[-0.0019, 0.0020]$; the paired median is exactly zero. At each error setting separately, the source-cluster interval includes zero.

More strongly, the magnitude of interior expansion does not predict the epistemic contrast. Across the 312 pairs, the Spearman association between the AI/human log ratio in $\chi(P)$ and the paired change in mean theorem belief is $\rho_s = 0.046$ (95% source-cluster interval $[-0.07, 0.22]$;

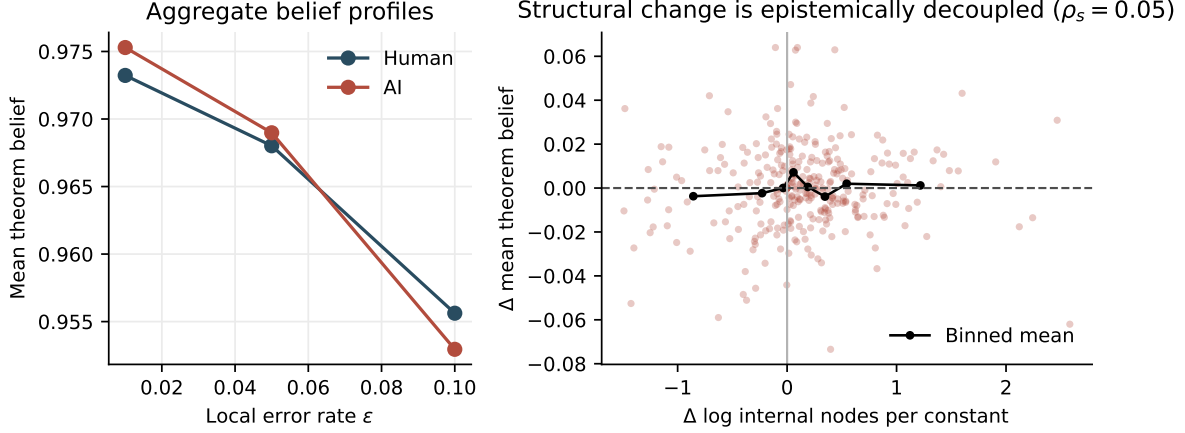


Figure 4: Epistemic decoupling in the proof-term model. (a) Mean theorem belief for human and AI terms at three symmetric local-error rates. (b) Pairwise structural change is nearly unrelated to pairwise theorem-belief change. The black line joins means of eight equally sized bins and is a visual summary, not a fitted causal model.

figure 4b). Proofs with much larger AI interiors are no more likely to receive lower modeled confidence.

This is the central extension of the phase-transition account. Viteri and DeDeo showed that networked proofs can sustain confidence despite local fallibility. The paired comparison shows a form of *representational robustness*: the same theorem can be realized by detectably different interiors without a corresponding shift in the modeled epistemic endpoint. Additional explicitness is not equivalent to an additional chain of independent liabilities.

5 Epistemic decoupling

We use *epistemic decoupling* for a specific empirical relation, not a general philosophical doctrine. For two proofs P_1 and P_2 of the same statement, an architectural statistic D and an epistemic functional B_ϵ are decoupled over a population when variation in $D(P_1, P_2)$ has little association with variation in $B_\epsilon(P_1) - B_\epsilon(P_2)$. Here D is interior load and B_ϵ is theorem belief in one network model. The result does not say that all presentations are equally understandable, that human readers are indifferent to structure, or that proof identity is exhausted by the conclusion.

The concept nevertheless clarifies three issues.

5.1 Cognitive units are not kernel nodes

Experts do not normally validate a proof by assigning equal attention to every primitive expression. They move backward from goals, inspect warrants selectively, reconstruct omitted steps, test examples, and use familiar lemmas as chunks (Weber, 2008; Mejía-Ramos and Weber, 2014; Panse et al., 2018). The stable modeled endpoint therefore creates a productive tension. If large term-level differences do not alter B_ϵ , then the decisive cognitive variation may occur at the mapping from term nodes to reader-visible units. A named lemma, an explicit intermediate claim, and a compiler-generated coercion should not receive the same prior, attention, or error model.

The interface-interior distinction offers a route forward. Constants and named local claims are candidate attention anchors. Interior subgraphs can be chunked beneath them. A cognitive model

can then treat proof reading as adaptive navigation across scales rather than uniform diffusion over a fixed graph. This extends the phase-transition framework from the topology of a certificate to the interaction between certificate and reader.

5.2 Correctness, correspondence, and explanation

A formal proof establishes that a term inhabits a type, conditional on the formal environment. It does not alone establish that the statement faithfully represents the intended mathematics or that the term tracks the reasons expressed by an informal proof (DeDeo and Duede, *forthcoming*). Nor does it guarantee explanation. Proofs can convey methods, dependencies, and unifying ideas beyond their conclusions (Rav, 1999; Mancosu, 2008).

Our data add a multiplicity problem: even after the statement is fixed, many checked interiors are possible. Which one should count as the mathematical proof for purposes of understanding, teaching, citation, or repair? Epistemic decoupling means that modeled confidence cannot by itself select among them. Other virtues—explanatory relevance, modularity, stability under library change, and capacity for communal correction—must enter. This conclusion is compatible with fallibilist accounts on which the epistemic achievement of mathematics includes not only avoiding error but organizing practices so that errors can be found and repaired (De Toffoli, 2021).

5.3 From proof search to proof design

Current AI-for-mathematics systems are optimized primarily to find any term that passes the kernel. That objective is extraordinarily valuable, but it is indifferent among many proofs of the same statement. The paired results suggest that this indifference is already visible in proof architecture: AI systems often add scaffolding behind a relatively stable interface.

The engineering response should not be to minimize length indiscriminately. A longer proof can be better decomposed, and a shorter proof can hide crucial ideas in automation. The relevant task is *proof design*: choose intermediate claims, names, dependencies, and abstraction boundaries so that a valid certificate becomes a useful mathematical artifact. Recent work on agentic proof refactoring explicitly targets readability, modularity, maintainability, and reuse rather than compilation alone (Fu et al., 2026). Human–AI formalization studies likewise indicate that users value high-level control and flexible collaboration (Collins et al., 2026). Our interface measure supplies one structural language for studying these goals, but only downstream human and library outcomes can validate it.

6 A decisive next experiment: from certificates to readers

The present study compares naturally occurring proof pairs and evaluates a theoretical belief model. The next study should intervene on proof architecture while correctness and mathematical content are held fixed.

Select 40–60 Lean theorems spanning algebra, number theory, combinatorics, and analysis. For each theorem construct four kernel-equivalent presentations:

1. the original human-provenance proof;
2. a raw AI-generated proof;
3. a refactored AI proof with named intermediate lemmas and an intentionally designed interface;
4. a mechanically expanded human proof that adds internal steps without changing its conceptual plan.

Independently randomize the displayed authorship label. This separates actual production, structural expansion, refactoring, and social expectations.

Recruit advanced students and professional mathematicians for complementary tasks: validate the proof, identify a seeded defect in an informal rendering, summarize the main idea, predict which assumptions are essential, transfer the method to a nearby theorem, and repair the proof after a small API change. Record time, eye movements or navigation traces, confidence trajectories, and which interface elements are inspected. The primary outcomes should be preregistered and sparse: validation accuracy, calibrated confidence, conceptual recall, transfer, and repair cost.

A multilevel model can then separate theorem, participant, presentation, and provenance-label effects. The key mechanistic test is whether interface design mediates any effect of interior load on comprehension and repair. The Viteri–DeDeo dynamics can be fit to within-reader confidence trajectories after introducing node-specific priors and attention weights. A successful extension would predict not only final belief but where readers look, where doubt persists, and when a proof is chunked into a trusted module.

A parallel longitudinal study should follow proofs after they enter a library. Outcomes include later citation, survival through API changes, reuse of introduced lemmas, reviewer effort, and the number of corrective commits. This would test whether a well-designed interface converts a certificate into communal mathematical infrastructure. It would also provide a behavioral and social criterion for artificial mathematical intelligence: not merely producing correct proofs, but participating in the practices through which proofs become understandable, revisable, and reusable (Pantsar, 2026).

7 Limitations

The paired design removes theorem identity as a confound, but it does not identify a pure causal effect of artificial authorship. “Human” and “AI” are dataset provenance labels; editing, retrieval, shared training data, and proof-selection pipelines can blur the categories. The corpus is dominated by competition mathematics and the 12 source groups are coarse.

The source interface measures are lexical heuristics. The term interface is more authoritative but remains representation-dependent: constants are counted at the root-term frontier, and different elaborator or Mathlib versions can change expression structure. Pairwise exclusion of error-recovery terms removes 36.8% of eligible pairs and may select on proof complexity. Our interior-load ratio is therefore a proposed measurement construct, not a universal ontology of proof.

Finally, the belief result is model-internal. Thirty-two Monte Carlo chains reduce simulation noise but cannot establish that the priors, node granularity, update dynamics, or symmetric error rates describe human proof reading. The near-zero contrast is best read diagnostically: within this model and representation, larger AI interiors do not create an epistemic deficit. Human behavior could reveal strong differences at the presentation layer that the term graph omits.

8 Conclusion

Holding the theorem fixed reveals both continuity and difference. Human and AI proofs share a strong theorem-imposed architecture, yet current AI proofs place more local and elaborated structure behind each named interface element. That interior expansion does not reduce theorem confidence in the network dynamics that motivated the original theory of epistemic phase transitions, nor does its magnitude predict the human–AI confidence contrast.

The result advances the study of mathematical belief from a question about whether proof networks can sustain confidence to a question about which features of a proof must remain stable for confidence, understanding, and reuse to survive changes in representation. In an era of abundant formal certificates, the scientific object is no longer the derivation alone. It is the coupled system of theorem, certificate, interface, reader, and library.

A Supplementary estimates and reproducibility

Table 1: Principal paired estimates. Difference rows report AI minus human; ratio rows report AI divided by human. All intervals resample the 12 problem-source groups.

Measure	Human median	AI median	Paired estimate	95% interval
Script lines	23	26	+2	[1, 3]
Explicit have claims	3	6	+1	[0, 1]
Loose source premise proxy	5	4	0	[0, 0]
Strict dotted premise proxy	3	2	0	[0, 0]
Local claims per tactic	.30	.50	1.41×	[1.27, 1.53]
Elaborated term nodes	3,816.5	4,561	+290	[20, 529]
Distinct elaborated constants	200.5	199	−1	[−3, 0]
Constant-node share	.0530	.0456	−.0046	[−.0061, −.0028]
Internal nodes per constant	17.79	20.84	1.12×	[1.06, 1.20]
Mean theorem belief, three settings	—	—	+ .00012	[−.0019, .0020]

The analysis script is `code/standalone_paper_analysis.py`; its machine-readable summary is `results/standalone_paper/summary.json`. It uses the corrected paired source table, the clean paired `term0` table, and the 32-run belief output. The 20,000-resample analysis was executed both locally and as ORCHARD Slurm job 128546; the resulting summaries were byte-identical. The ORCHARD output archive has SHA-256 `b90b31e99003d40ffdc9e8bad9a027d00d6302cc2798d7c4cf9a520468de48c4`.

The compact derived tables and code are included with this working directory. The raw elaborated graphs and upstream corpus should be deposited with immutable manifests and license metadata before publication so that extraction can be reproduced end to end.

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